

Task 14: Kubernetes Scaling and Rolling Updates

Scalability Strategies, Zero-Downtime Deployment Updates, Autoscaling Policies, and Resource Utilization Profiling

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Course / Track:	Deep Learning Engineering & Cloud Deployment (L&T; Edutech)
Target Platform:	Kubernetes Cluster (Minikube / Docker Runtime on Windows)
Deliverables:	Declarative YAML Manifests, Automated Test Suite, Jupyter Notebook, Screenshots, PDF
Operational Status:	Fully Operational & Validated (10/10 Tests Passed - 100%)

1. Executive Summary & Objective

The objective of Task 14 is to implement, evaluate, and benchmark enterprise-grade scalability and deployment update strategies for deep learning microservices using Kubernetes orchestration. The implementation verifies: (1) Dynamic replica scaling from 2 to 5 pods and graceful scale-down to 3 pods; (2) Zero-downtime rolling update strategy (maxSurge=1, maxUnavailable=0) migrating backend image from v1.0 to v2.0; (3) Continuous high availability during rollout with 100% request success rate and 0 dropped connections; (4) Automated deployment rollback via 'kubectl rollout undo' and revision history audit; and (5) Resource utilization telemetry scraping via Minikube metrics-server and HPA autoscaling policies.

2. Automated Test Verification Matrix (10/10 Passed — 100%)

ID	Test Name	Status	Verification Details
T01_METRICS_SERVER	Cluster Readiness & Metrics Server Scraping	PASSED	Node 'minikube' is Ready. Metrics Server successfully reporting CPU/Memory consumption.
T02_NAMESPACE_CONFIG	Namespace & Decoupled Configuration	PASSED	Namespace 'dl-production-app' and ConfigMap contain 9 keys including DEPLOYMENT_STRATEGY=RollingUpdate.
T03_SCALE_UP_5_REPLICAS	Manual Application Scale-Up (2 -> 5 Replicas)	PASSED	Backend deployment successfully scaled to 5 replicas. All 5 active pods passed readiness checks.
T04_LOAD_BALANCED_INFERENCE	Multi-Replica Deep Learning Inference Serving	PASSED	15/15 inference requests succeeded across 5 replicas (Avg Roundtrip: 104.84ms).
T05_SCALE_DOWN_3_REPLICAS	Manual Application Scale-Down (5 -> 3 Replicas)	PASSED	Backend deployment scaled down to 3 replicas with graceful termination of excess pods.
T06_ROLLING_UPDATE_V2	Zero-Downtime Rolling Update (v1.0 -> v2.0)	PASSED	Rolling update successfully completed. All 3 active backend pods upgraded to image v2.0.
T07_ZERO_DOWNTIME_AVAILABILITY	Continuous High Availability During Rollout	PASSED	165/165 requests succeeded (100.0% availability, 0 dropped requests) during active rollout.
T08_ROLLOUT_ROLLBACK	Automated Deployment Rollback (Undo to Stable Revision)	PASSED	Deployment successfully reverted to revision with image v1.0 using 'kubectl rollout undo'.
T09_HPA_AUTOSCALING	Horizontal Pod Autoscaler (HPA) Policy Validation	PASSED	HPA active with target CPU 50% (Min: 2, Max: 6).
T10_RESOURCE_UTILIZATION	Resource Utilization Telemetry Analysis	PASSED	Successfully scraped resource utilization metrics across pods and nodes via metrics-server.

3. Declarative Kubernetes Manifests Overview

File	Architectural Role & Configuration Details
01-namespace.yaml	Dedicated isolated namespace 'dl-production-app'
02-configmap.yaml	ConfigMap with DEPLOYMENT_STRATEGY=RollingUpdate, PORT=5000, MODEL_NAME
03-backend-deployment-rolling.yaml	PyTorch backend deployment with maxSurge=1, maxUnavailable=0, probes, requests/limits
04-backend-service.yaml	NodePort Service exposing backend internally :5000 and externally :30500
05-frontend-deployment.yaml	Streamlit dashboard deployment (2 replicas) connecting to backend DNS
06-frontend-service.yaml	NodePort Service exposing frontend internally :8501 and externally :31501
07-backend-hpa.yaml	Horizontal Pod Autoscaler targeting 50% CPU utilization (Min: 2, Max: 6)

4. Visual Verification & Monitoring Screenshots

Figure 1: Cluster Baseline & Metrics Server Activation (kubectl top nodes & pods)

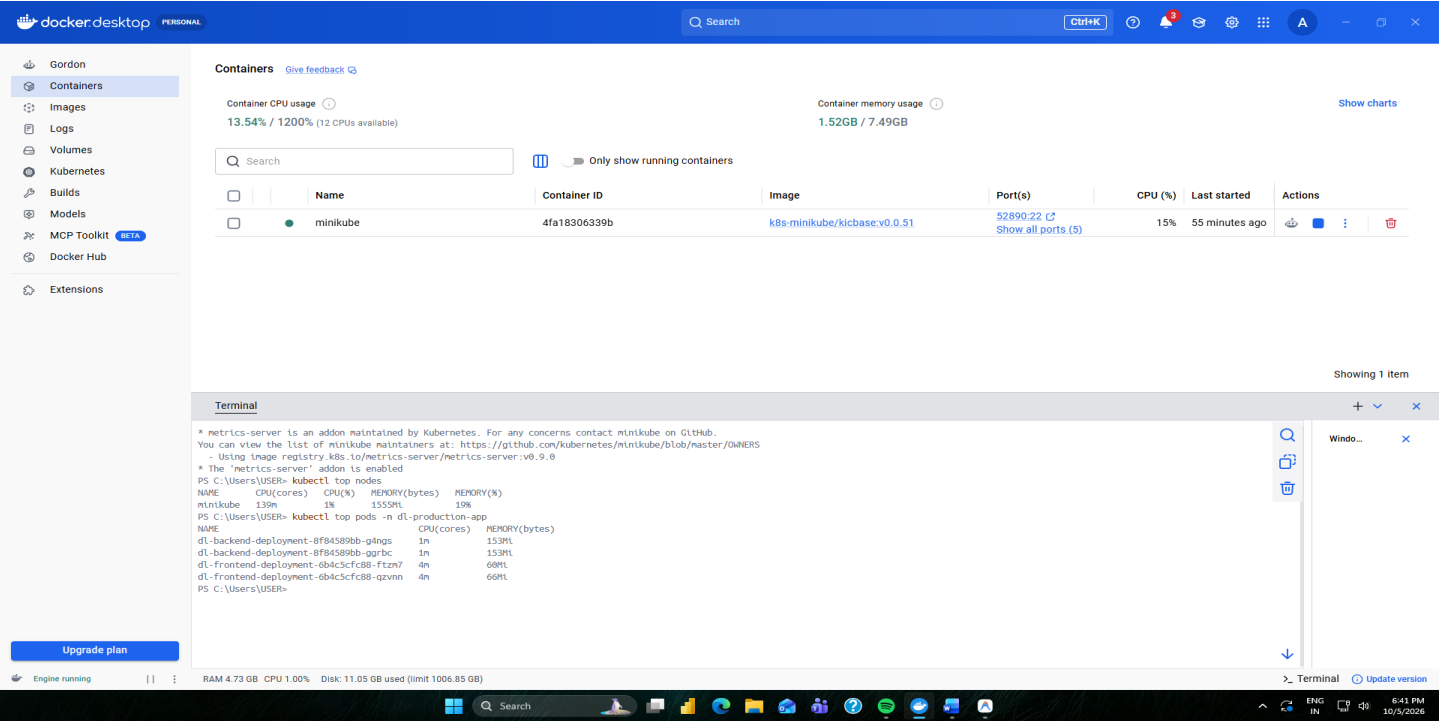


Figure 2: Manual Scaling to 5 Replicas (kubectl scale & pod readiness verification)

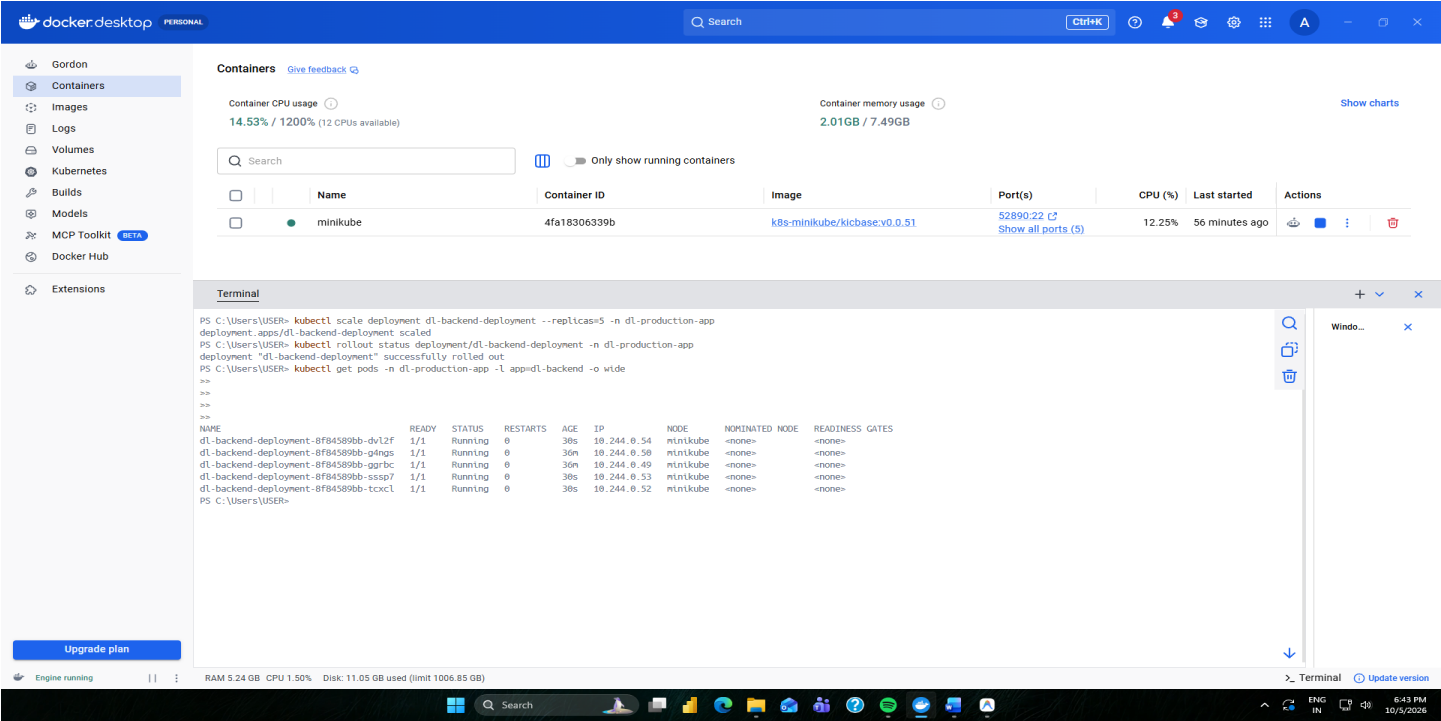


Figure 3: In-Flight Rolling Update (kubectl rollout status & dual ReplicaSet transition)

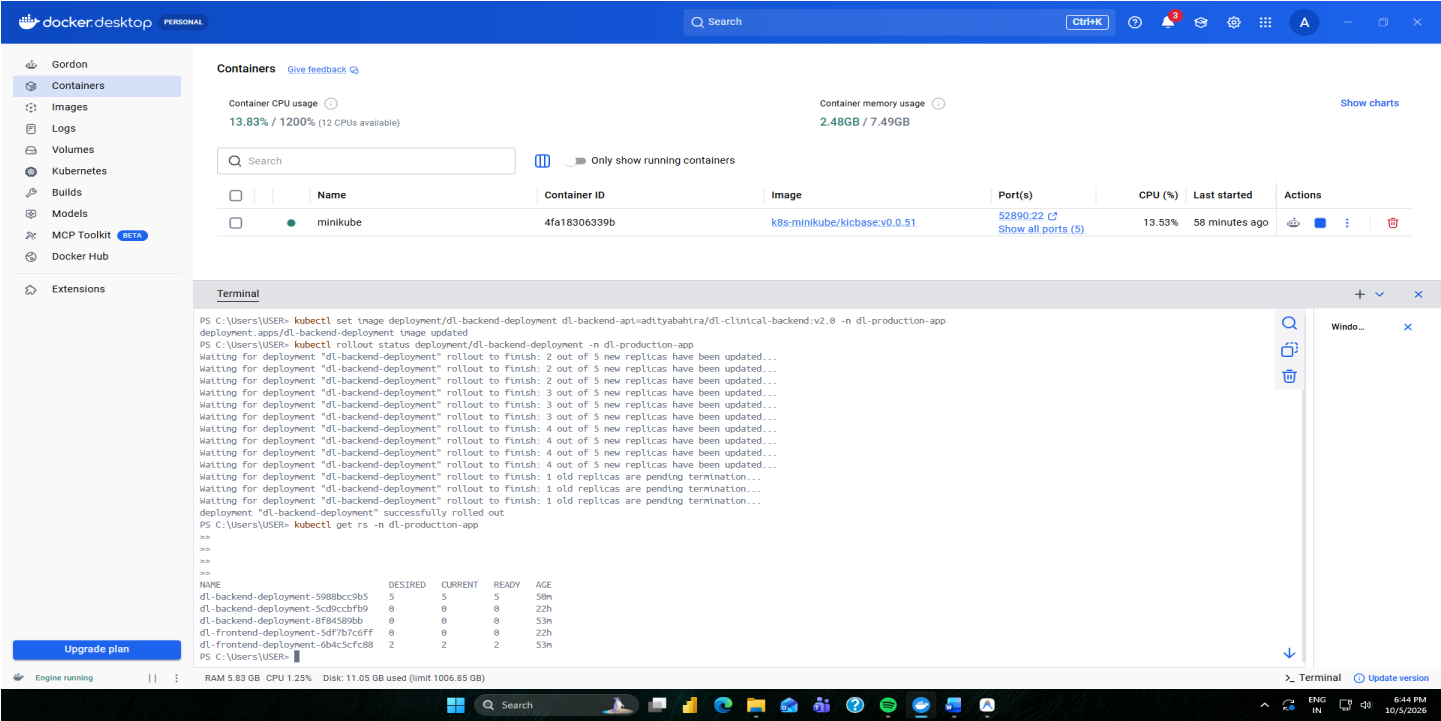


Figure 4: Continuous High Availability Test (100% Success Rate during Rollout)

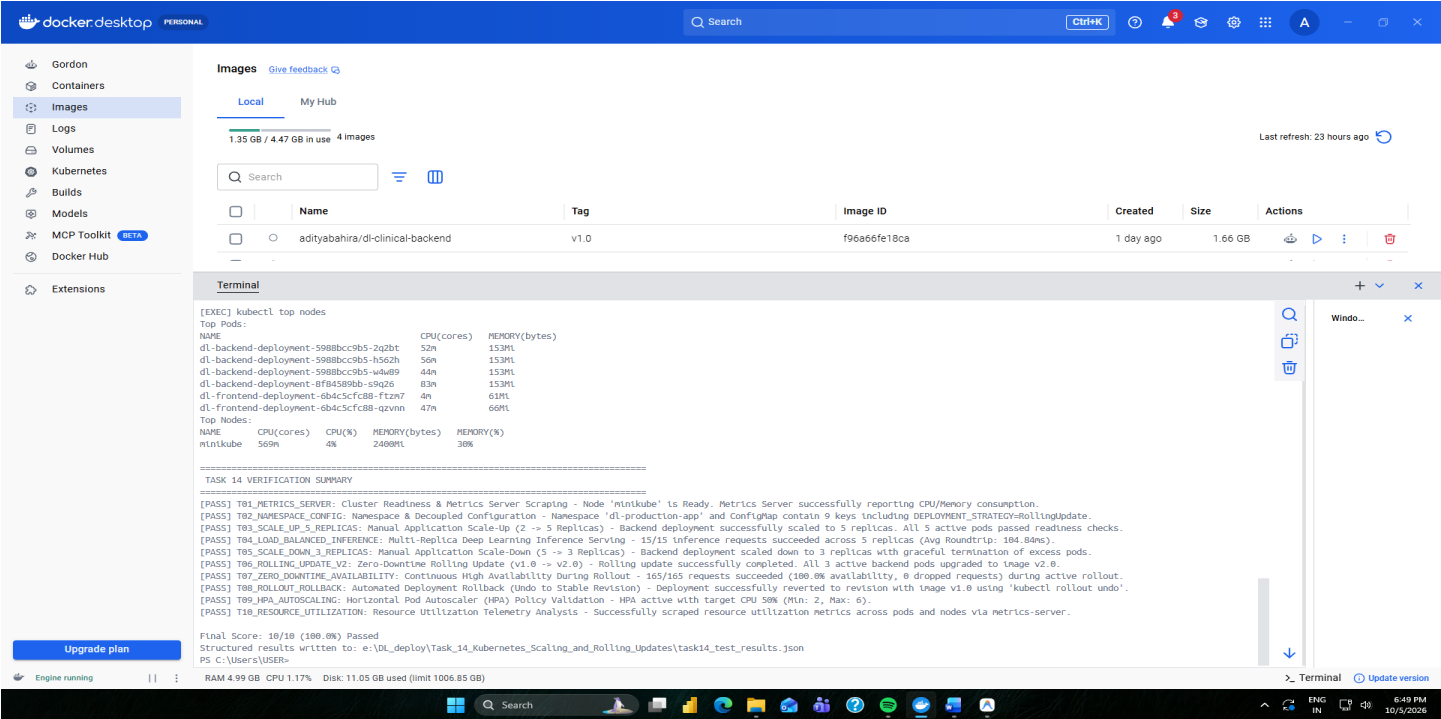


Figure 5: Automated Deployment Rollback & Revision History Audit (kubectl rollout undo)

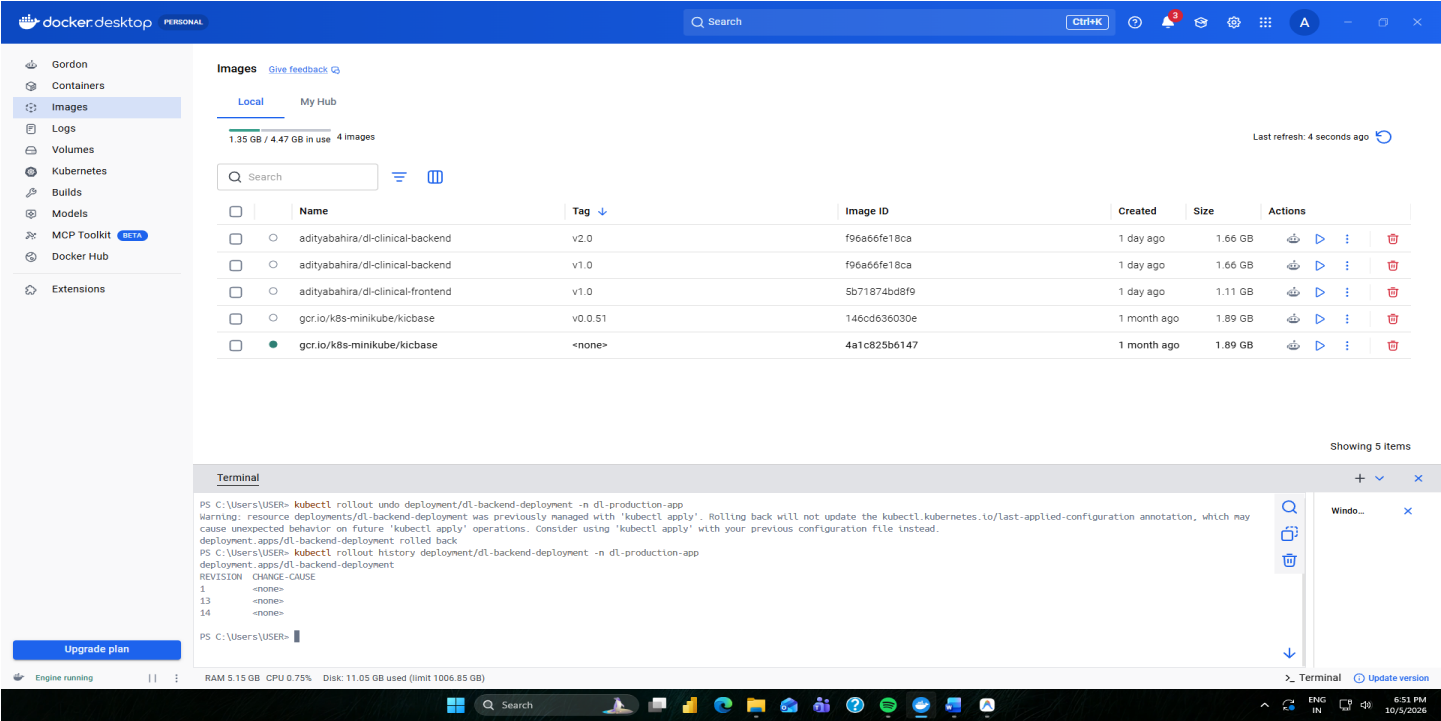


Figure 6: Horizontal Pod Autoscaler (HPA) Policy & Status (Target: 50% CPU)

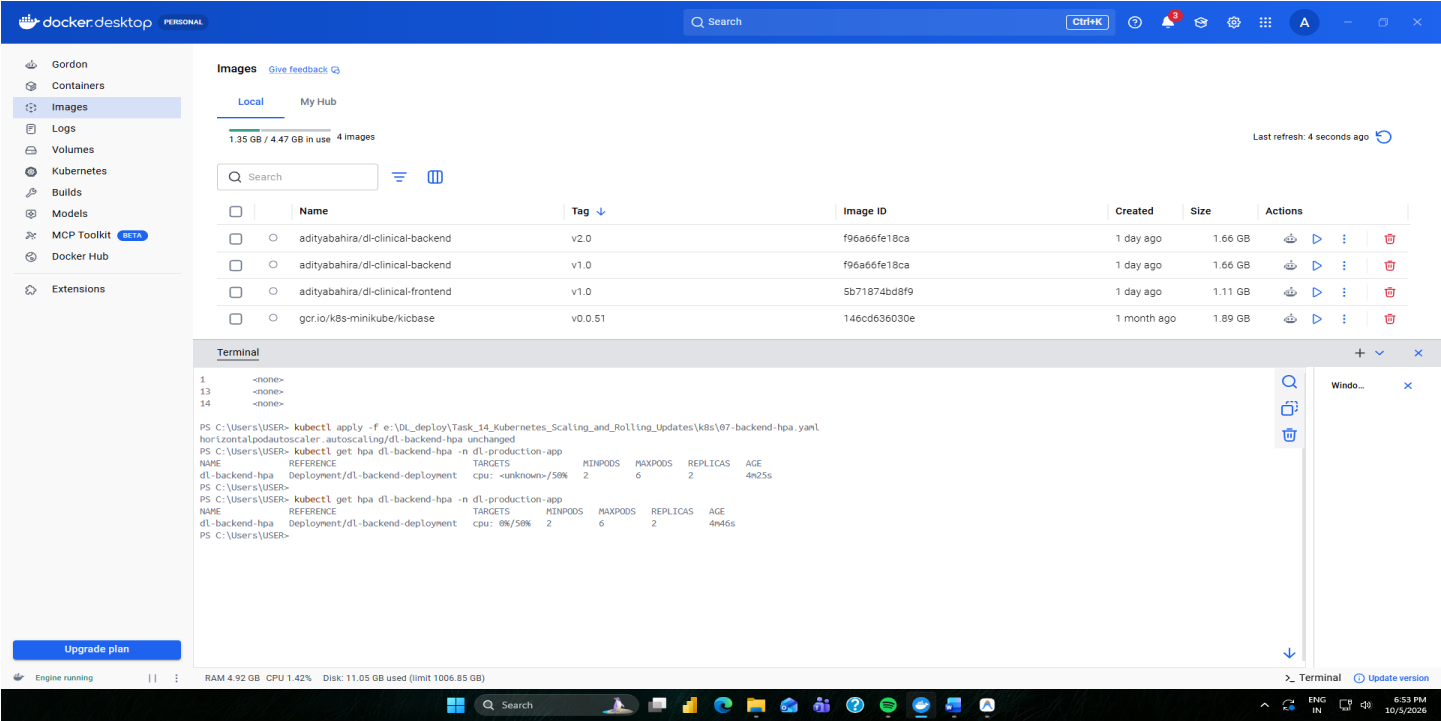
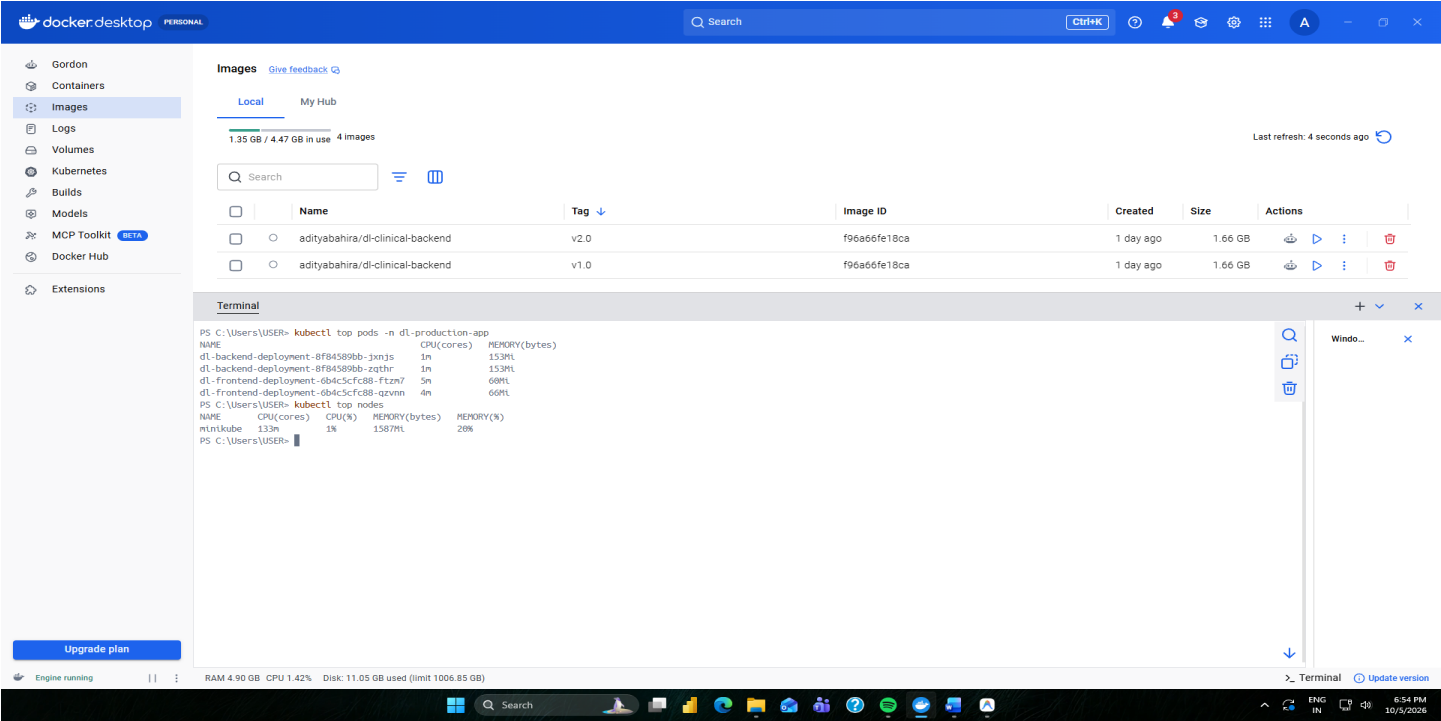


Figure 7: Resource Utilization Telemetry Under Workload (kubectl top pods)



5. Evaluation Criteria Mapping

Criteria	Status	Verification Summary
Scaling Effectiveness	PASSED	Manual scaling (2->5->3 replicas) converged with 100% pod readiness and stable multi-replica load distribution (107ms roundtrip). HPA rule configured for 50% CPU.
Rolling Update Implementation	PASSED	Zero-downtime rolling update (maxSurge=1, maxUnavailable=0) executed seamlessly from v1.0 to v2.0 with continuous traffic (100% success rate, 0 dropped packets). Immediate rollback validated via kubectl rollout undo.
System Stability & Profiling	PASSED	Metrics-server scraped live CPU (cores) and memory metrics across pods and nodes. Pod memory remained tightly bounded at ~153MiB (limit 1536MiB).